

Coursework Approval Slip – Dataset Provenance & Specification

Course: Quantitative Data Analysis (M1 – S7) **Course Code:** 2627_EC0_2_EN_009
– Academic Year 2026–2027 **Institution:** EM Normandie Business School **Lecturer:**
Dr. NGUYEN Anh-Tuan **Grade Weight: 10 Marks (Provenance & Dataset Quality)**
Submission Milestone: Session 1 Approval Gate (Slide 35)

1. Administrative Identification

Field	Description / Value
Team Members	David & Group Consulting Team (M1 S7 PGE)
Programme & Campus	Programme Grande École – EM Normandie
Dataset Title	S&P 500 Corporate Governance & Financial Performance Dataset (FY2025)
Source URL 1 (Constituents)	Wikipedia S&P 500 Directory (sourced from S&P Dow Jones Indices)
Source URL 2 (Fundamentals)	Yahoo Finance (sourced from SEC 10-K audited annual filings)
Source URL 3 (Governance Risk)	Institutional Shareholder Services (ISS) via Yahoo Finance API
Producer & Year	Standard & Poor's, ISS Governance, and Yahoo Finance / SEC filings (Audited FY2025, ratings Q3/Q4 2025)
Downloadable File Format	.csv (sp500_esg_cleaned.csv / sp500_esg_jamovi.csv) and .xlsx (sp500_esg_jamovi.xlsx)

2. Dataset Structural Compliance (All Boxes Ticked)

Requirement (Slide 29)	Project Specification	Compliance
One row = one...	One row = one publicly traded S&P 500 corporation	 YES
N (Observations)	$N = 503$ firms (exceeds the ≥ 100 minimum threshold)	 YES
Nominal Variable (2–5 categories)	Sector — exactly 5 consolidated categories : 1. <i>Industrials & Energy</i> ($n = 148$, 29.4%, Mode) 2. <i>Consumer</i> ($n = 116$, 23.1%) 3. <i>Tech & Comms</i> ($n = 108$, 21.5%) 4. <i>Finance</i> ($n = 71$, 14.1%) 5. <i>Healthcare</i> ($n = 60$, 11.9%)	 YES
Ordinal Variable (≥ 1)	1. Governance_Risk_Level (3 ordered tiers: Low, Medium, High) 2. Market_Cap_Quartile (4 ordered tiers: Q1, Q2, Q3, Q4)	 YES
Continuous Variables (≥ 3)	6 continuous variables available: 1. Profit_Margin (Net profit margin, FY2025) 2. ROE (Return on Equity, FY2025) 3. Market_Cap_B (Market capitalization in \$ Billion) 4. Total_Revenue_B (Annual turnover in \$ Billion) 5. Beta (Systematic risk coefficient, 5Y monthly) 6. Overall_Governance_Risk (ISS QualityScore, 1 to 10 scale)	 YES
Raw & Authentic Data	Live extracted from financial APIs; zero pre-processed, synthetic, or beginner datasets (Iris, Titanic, Boston Housing strictly excluded).	 YES

3. Managerial Research Question (Problematic)

“Does corporate governance risk (ISS QualityScore) compromise financial profitability and market valuation among S&P 500 firms, or does strong governance provide a resilience and performance premium?”

Alignment with the 4 Course Sessions:

- **Session 1 (Univariate):** What are the distribution profiles, central tendencies, and dispersions of profitability and governance risk? (Skewness, kurtosis, normality checks).
 - **Session 2 (Bivariate):** Are high-risk governance firms disproportionately concentrated in certain sectors or size quartiles (Chi-Square)? Is governance risk correlated with profit margins (Pearson r)?
 - **Session 3 (ANOVA):** Does profitability or governance risk differ significantly across the 5 economic sectors (One-Way ANOVA & Games-Howell/Tukey HSD)?
 - **Session 4 (Regression):** Does governance risk predict profit margins when controlling for firm size (Market_Cap_B) and market volatility (Beta)?
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4. Lecturer Validation Gate

Lecturer Name: Dr. NGUYEN Anh-Tuan

Approval Decision: ☐ APPROVED (10 / 10 marks awarded)
☐ REVISION REQUIRED

Signature: _____

Date: ____ / ____ / 2026